Two-Primary Birch–Swinnerton-Dyer: Analytic Certificates, Index Squares, and Determinant Defects

Dr. Muralidhara Subbarao
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www.integralresearch.net
Stony Brook University
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Abstract

We assess the applicability of selected Ram-calculus constructions to the manuscript *The two-primary Birch–Swinnerton-Dyer formula in Selmer corank at most one*. The source claims an exact valuation formula at two for every rational elliptic curve of usual two-power Selmer corank at most one. We map its complete section structure while reading selected normalization, determinant, rank-detection, and final-deduction chains; we do not independently certify its integral arithmetic proof. Starting from actual private proof texts, we specialize the Mellin moments to closed low-rank kernels, prove explicit exponential-fit panel bounds, and replace full saturation by a two-primary index certificate for the valuation task. At a known simple split zero, we retain a determinant quotient defect and derive its exact signed contribution to the sum of BSD discrepancies. Conditional rational reconstruction states the additional arithmetic information that real interval calculations need. All real components, the source's height convention, full torsion, and index squares are retained. Contribution assessment: C2/5, useful local certification and bookkeeping refinements are proved, but the deep integral comparisons and the universal two-primary theorem remain imported.

1 Scope and outcome

This is report M004, applicability-catalogue id 4, family 2. Applicability **L3** means that the selected analytic and lattice steps admit useful transfers. It is distinct from the realized-contribution score **C2**: a little contribution, not a significant advance on the source's arithmetic theorem. The local statements below are proved here; their elementary or classical ingredients are not claims of historical novelty. No numerical speedup, new elliptic-curve BSD instance, or unconditional new Selmer converse is asserted.

The pinned build contains 17 text files, 635,589 bytes, and 9,525 lines. The literal-read ledger records 2,850 distinct lines across those files. All section and subsection headings were mapped. The complete build driver, bibliography, introduction, classical-normalization section, positive-completion section, and conclusion were read. Other files were read selectively. Two actual private originals contribute 789 distinct lines: 705 in `TR-item-004-astra.tex` and 84 in the revised foundation report `TR-rev-1-old-tr250.tex`. Neither original was read in full, and no whole-corpus reading is claimed.

The source pin is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Parent-provided batch-35 evidence authenticates the M004 subtree as unchanged at the captured October 8 target `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. This is byte identity against captured Git trees, not a fresh live-branch proof check. We use a report-specific snapshot of that evidence. The parent's seven exact finite normalization tests are supplementary evidence, not author tests or arithmetic-proof verification.

Three levels of assertion. *Source* denotes a theorem or comparison claimed by the pinned manuscript, with its uninspected prerequisites retained. *Local* denotes a deduction with a proof in this report. *Conditional* denotes a local deduction requiring separately supplied arithmetic certificates. Reading a displayed proof is not verification of every imported comparison, limit, cup product, or integral model used by it.

2 Exact arithmetic and analytic objects

Let $E/\mathbb{Q}$ be an elliptic curve with nonzero discriminant, a global minimal Weierstrass model, conductor N, and minimal Néron differential ω_E. Put $T_E = E(\mathbb{Q})_{\mathrm{tors}}$ and $\Lambda_E = E(\mathbb{Q})/T_E$, a free abelian group. Throughout,

$$s_2(E) = \operatorname{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty}(E/\mathbb{Q}), \qquad v_2(2) = 1.$$

The Selmer group uses the ordinary local Kummer conditions at *every* place, including the real place; no strict or positive variant is silently substituted.

The period and height conventions are exactly the source's:

$$\Omega_E = \int_{E(\mathbb{R})} |\omega_E| = c_\infty \Omega_0, \qquad c_\infty = \#\pi_0 E(\mathbb{R}) \in \{1, 2\}, \tag{2.1}$$

$$H(P) = \lim_{j\to\infty} 4^{-j} h_x([2^j]P), \qquad B(P,Q) = \tfrac{1}{2}(H(P+Q) - H(P) - H(Q)). \tag{2.2}$$

Here $h_x(P) = \log\max(|a|,|b|)$ for reduced $x(P) = a/b$, and $h_x(O) = 0$. Thus $B(P,P) = H(P)$; H is attached to $2[O]$. The height attached to $[O]$ is $H/2$. We preserve this convention, rather than diagnosing a factor-of-two error merely from a different software convention. The pairing is positive definite on $\Lambda_E \otimes \mathbb{R}$, and Reg_E is its determinant on a full integral basis, with empty determinant 1 in rank zero. Write $C_E = \prod_{\ell \text{ finite}} c_\ell(E)$.

Source input: For every $E/\mathbb{Q}$ with $s_2(E) \le 1$, the manuscript claims

$$r = \operatorname{rank} E(\mathbb{Q}) = \operatorname{ord}_{s=1} L(E,s) = s_2(E), \qquad \#\,\mathrm{Sha}(E/\mathbb{Q}) < \infty, \tag{2.3}$$

$$A_E = \frac{L^{(r)}(E,1)}{r!}, \qquad Q_E = \frac{A_E (\#T_E)^2}{\Omega_E \operatorname{Reg}_E C_E} \in \mathbb{Q}_{>0}, \qquad v_2(Q_E) = v_2(\#\,\mathrm{Sha}(E/\mathbb{Q})). \tag{2.4}$$

There are no residual-image or reduction-at-two exclusions in this *source statement*. The present report does not establish it anew. It asserts neither $Q_E = \#\,\mathrm{Sha}(E/\mathbb{Q})$ as a rational equality nor the other prime parts; equal two-adic valuations do not imply equal rationals.

The analytic space. For the real-coefficient normalized newform $f_E \in S_2(\Gamma_0(N))$, set $\alpha = 2\pi/\sqrt{N}$, $g(x) = f_E(ix/\sqrt{N})$ for $x \ge 1$. Its Fourier expansion $g(x) = \sum_{n\ge1} a_n e^{-\alpha n x}$ has exact integer coefficients and the standard envelope $|a_n| \le d(n)\sqrt{n} \le 2n$. For every fixed $k \ge 0$, g belongs to $L^1([1,\infty), (1+\log^k x)\,\mathrm{d}x)$; stronger polynomially weighted integrability follows from exponential

decay. Mellin differentiation is justified locally uniformly in the complex parameter. There is no physical PDE boundary condition or gauge: the relevant constraint is the functional equation. Finite panels use C^{q+1} functions or finite polynomial spaces; point jets are not asserted continuous on arbitrary L^2 data.

3 Whole-source structure and dependency map

The following is a map of every build section file, not a claim that every proof in it was inspected. File 12 contains two numbered sections; file prefixes, rather than inferred printed section numbers, are used below. Exact line intervals and hashes are in the accompanying ledger.

File	Structure and role	Read scope and imported boundary
01	Introduction: statement, normalization, history, three comparisons, proof order.	Full 329 lines. Source assertions are distinguished from their later proof inputs.
02	Classical facts: low-rank inputs and parity; absolute Gross–Zagier heights; split discrepancy.	Full 173 lines. Modularity, duality, parity, nonvanishing and Gross–Zagier–Kolyvagin are imported classical theorems.
03	Finite marked cochains, localization/duality, evaluations, determinant volumes, line/square switches, Kummer volumes, bounded-series tests.	Lines 207–236 and 321–416. Determinant and central-volume arithmetic examined; construction and exact compatibility of the finite models are not audited.
04	Relative-symbol lattice, smoothed Siegel classes, reciprocity, norm relations, period law, positive twists.	Lines 572–734. Unnormalized period-law proof read; earlier integral lattice, reciprocity and norm constructions remain inputs.
05	Positive determinant: parameter ring, residual concentration, horizontal switches, fixed-divisor removal, rank-zero center.	Lines 1–78 and 499–558. Integral determinant statement and central calculation read; residual concentration and pole removal imported.
06	Tame logarithmic comparison: direction/cochains, biextensions, holomorphic kernel, discriminant-prime trace, spectral evaluation, intersections, completion.	Lines 1–147. Exact logarithm identity and rank-free torsion test read as statements; the long analytic/intersection proof is not inspected.
07	Positive completion: rank-one center, compatible direction, nonzero Bockstein, analytic-zero detection.	Full 408 lines. Logical chain examined, but its finite-stage realization, compatible limits and earlier spectral identity remain deep inputs.
08	Strict split determinant, transverse parameter, depleted primitives, disk measures, ring-class interpolation, residual primitivity.	Lines 1–89. Reducible setting, strict model and residual-order calculation read; measure construction and interpolation not audited.
09	Odd and dyadic local conditions, inert derivative identities, horizontal divisibility/unit quotient, central rank/exact factor, Heegner rank test.	Lines 1–13 and 348–436. Center and exact-factor chain read; local membership, logarithm lattices and unit theorem imported.
10	Reducible seed: local-condition labels, full rational plane, one-line seed, split imaginary partner.	Lines 1–37. Finite-seed statement and its logical role read; descent and prime-selection proofs not inspected.
11	CM realizations, elliptic units, logarithms, known low-rank formulas, paired determinant, horizontal divisors, corank-one converse.	Lines 1–129. Realizations and period normalization read; elliptic-unit integrality, paired formula and converse remain imported.
12	Conjugate Bockstein alternation, then imaginary S_3: comparator/seed, paired systems, central switches, residual concentration, divisibility, center.	Lines 1–99. Alternation proof and initial seed/comparator read; integral stage hypotheses and later determinant transfer not independently checked.

File	Structure and role	Read scope and imported boundary
13	Real S_3: level, real-component functional, integral Hecke module, Pfaffians, cofactor/product divisibility, horizontal logarithms, clearing factor, global specialization.	Lines 1–58 and 134–264. Detector proof read, with earlier setup only partially inspected; Hecke/Pfaffian integral construction and its specialization remain imported.
14	Cyclic cubic: Selmer seed, regularity/divisibility, primitive ray coordinate, log lattice, comparator disk centers, anchor.	Lines 1–118. Seed and comparator setup read; ray-unit primitivity and exact integral comparison not audited.
15	Assembly of residual cases, main theorem, signed-squarefree density corollary.	Full 129 lines. Final deductions checked conditional on the preceding propositions and the separate density input.

The 79-line build driver includes these 15 files and the 395-line bibliography; both were fully read. Thus every assigned file is mapped and has a literal read, but 6,675 source lines remain outside that read.

3.1 The final deduction, with hypotheses visible

In known analytic ranks at most one define

$$X(A) = v_2(Q_A/\#\,\mathrm{Sha}(A/\mathbb{Q})).$$

The classical low-analytic-rank theorem and rationality comparisons make this defined. The source's three assertions have these roles:

1. For non-CM A of known analytic rank at most one, $X(A) \geq 0$. If a positive fundamental discriminant a, prime to $2N_A$ (or $a = 1$), has analytic rank at most one and $X(A^a) = 0$, then $s_2(A) \leq 1$ implies analytic rank $s_2(A)$ and $X(A) = 0$.

2. A split-pair anchor supplies coprime odd fundamental h, k, allowing $h = 1$, with $k < 0$, the prescribed splitting, and $\mathrm{an}(A^h) + \mathrm{an}(A^{hk}) = 1$, $X(A^h) + X(A^{hk}) = 0$.

3. The CM comparison proves the corresponding assertion directly for CM curves of $s_2 \leq 1$.

Local result: Conditional assembly. In the non-CM case both anchor ranks are at most one. Nonnegativity forces both discrepancies to vanish. Exactly one of h, hk is positive. Because h and k are coprime odd fundamental discriminants (allowing $h = 1$), hk has the required discriminant type. Positive transfer applies to the original curve. The classical rank/finiteness theorem then supplies the full finite Sha. This short implication is checked; constructing the anchors is not replaced by it.

In particular, the source does not infer rank merely by detecting a two-Selmer seed. Its unknown rank-one center first uses a nonzero Selmer Bockstein and an integral unit, then the torsion implication of the tame spectral identity, to establish $L'(E, 1) \neq 0$. Only afterward does it use the known-rank central formula. This order is essential for the conditional extension in Section 8.

4 Public literature refresh and private proof routes

Public primary searches and opens were made on October 8, 2026, using public titles and public URLs only. No confidential source text, private title, or report passage was submitted. The latest pages that were accessible are evidence for their displayed claims, not certification of all their proofs.

Smith's primary record states a 50%/50% distribution of two-power Selmer coranks zero and one in every rational elliptic-curve quadratic twist family [4]. The displayed HTML introduction

formulates its counting theorem over all nonzero integer parameters and keeps the analytic Goldfeld conclusion conditional on BSD. We therefore do not silently identify that displayed sampling convention with the source's signed-squarefree convention. M004 explicitly imports the latter distribution from its separate Goldfeld companion; that companion has not been read here. The density consequence is not an output of finite Ram panels or the tests below.

Burungale–Tian's 2026 primary journal record gives the rank-zero p-converse for CM elliptic curves over $\mathbb{Q}$, for every prime p [5]. Burungale–Flach's primary abstract gives a full BSD theorem under stated CM, nonvanishing, and abelian torsion-field hypotheses [6]; these hypotheses must be retained before applying that result. Kriz–Li's institutional primary record reports congruences between Heegner logarithms and corresponding p-part formulas for explicitly constructed twists [7]. These records supply context, not the source's unrestricted non-CM two-primary theorem.

The author-hosted search result for Li–Tian–Yan–Zhu states a CM analytic-rank-one p-part formula for split p, requiring good ordinary reduction when $p = 2$ [8]. Direct PDF and DOI opens failed during this refresh; the claim is used only at the scope of that primary abstract snippet and the pinned bibliography, not as a newly read proof. The current public M004 GitHub PDF record was accessible [1]. We assess its locally pinned text and the parent's captured-tree byte identity, not a newly downloaded latest-branch manuscript. No public search outcome proves acceptance or independent verification of its universal claim.

Private route P1. The actual `TR-item-004-astra.tex` was read at lines 197–306, 395–803, and 866–1051. Its proved log-moment identity, coefficient and panel tails, fail-closed rank rule, height determinant bound, saturation gate, conditional discrete isolation, and revised prefactor treatment are directly relevant. The present report retains these safeguards and specializes them to $r = 0, 1$. Its older general-BSD status discussion is not used to dismiss a later dated source claim.

Private route P2. The actual revised foundation report was read at lines 121–181 and 455–477. Its finite moment representation $m = Ec$, invertibility/range requirements, and positive-weight polynomial unisolvence proof define the permissible use of "Ram coordinates" here. A finite coordinate change preserves a chosen function space; it does not create arithmetic integrality, an infinite-dimensional lower frame bound, or a smaller best-approximation error. Our weighted polynomial panel calculation is a concrete use of this representation principle.

5 Exact low-rank kernels and certified tails

Use the standard completion

$$\Lambda(E, s) = \alpha^{-s}\Gamma(s)L(E, s), \qquad \Lambda(E, s) = w_E\Lambda(E, 2 - s), \quad w_E \in \{1, -1\}.$$

The private Mellin route, by splitting at one and the Atkin–Lehner relation, proves

$$\Lambda(E, 1 + z) = \int_1^\infty g(x)(x^z + w_E x^{-z})\,\mathrm{d}x, \quad \Lambda^{(k)}(E, 1) = (1 + w_E(-1)^k)\int_1^\infty g(x)\log^k x\,\mathrm{d}x. \quad (5.1)$$

Since $\alpha^{-1}\Gamma(1) = \alpha^{-1}$, conversion of a *leading* derivative to L multiplies by α. This rule is not valid for an arbitrary higher jet with nonzero lower derivatives: then the product rule retains derivatives of $\alpha^s/\Gamma(s)$.

Theorem 5.1 (Low-rank exponential kernels)**.** *Let $E/\mathbb{Q}$ and its normalized modular form be as above. For $t > 0$ define $\mathrm{E}_1(t) = \int_t^\infty e^{-u}\,\mathrm{d}u/u$. If $w_E = 1$, then*

$$L(E,1) = 2\sum_{n\geq 1} \frac{a_n}{n} e^{-\alpha n}. \tag{5.2}$$

If $w_E = -1$, then $L(E,1) = 0$ exactly and

$$L'(E,1) = 2\sum_{n\geq 1} \frac{a_n}{n} \mathrm{E}_1(\alpha n). \tag{5.3}$$

For every integer $M \geq 1$, with $q = e^{-\alpha}$, the errors on truncating these sums at M are bounded respectively by

$$B_0(M) = \frac{4q^{M+1}}{1-q}, \qquad B_1(M) = \frac{4q^{M+1}}{\alpha(M+1)(1-q)}. \tag{5.4}$$

No rank conclusion is assumed in these identities.

Proof. Absolute convergence follows from $|a_n| \leq 2n$. Integrate the $k = 0$ moment termwise. For $k = 1$, integration by parts gives

$$\int_1^\infty e^{-\beta x} \log x\,\mathrm{d}x = \frac{\mathrm{E}_1(\beta)}{\beta} \qquad (\beta > 0).$$

In odd sign, lower vanishing removes the derivative of the completion factor; multiply the first completed derivative by α. This proves both formulas. The first tail is bounded by $4\sum_{n>M} q^n$. Since $\mathrm{E}_1(t) \leq e^{-t}/t$, the second is bounded by $4\alpha^{-1}\sum_{n>M} q^n/n$, which gives (5.4). □

This is a source-specific simplification of the general private central-jet machinery: rank zero needs no spatial quadrature, and rank one needs a single known special-function kernel. The tail bound is the low-order specialization of the revised private log-atom bound after the required completion prefactor, not a newly discovered estimate. Exact Fourier coefficients, the exact conductor and root number, and outward-rounded evaluations of α, exponentials and kernels remain necessary for a numerical certificate.

Proposition 5.2 (Signed kernel remainder)**.** *For every $t > 0$ and integer $K \geq 0$, let*

$$S_K(t) = e^{-t}\sum_{j=0}^{K-1} \frac{(-1)^j j!}{t^{j+1}}, \qquad R_K(t) = \mathrm{E}_1(t) - S_K(t).$$

The empty sum is zero. Then $(-1)^K R_K(t) \geq 0$ and

$$|R_K(t)| \leq \frac{e^{-t}K!}{t^{K+1}}, \qquad \frac{e^{-t}}{t+1} \leq \mathrm{E}_1(t) \leq \frac{e^{-t}}{t}. \tag{5.5}$$

Proof. Write $\mathrm{E}_1(t) = e^{-t}\int_0^\infty e^{-u}(t+u)^{-1}\,\mathrm{d}u$. The exact finite identity

$$\frac{1}{t+u} = \sum_{j=0}^{K-1} \frac{(-1)^j u^j}{t^{j+1}} + \frac{(-1)^K u^K}{t^K(t+u)}$$

gives the sign and, using $t + u \geq t$ and the gamma integral $K!$, the error bound. The upper bound takes $K = 0$. The lower bound is Jensen's inequality for the convex function $u \mapsto (t+u)^{-1}$ under the probability density e^{-u}, whose mean is one. □

This is a finite asymptotic enclosure, not a convergent power series for fixed t. Increasing K indefinitely can worsen it. Intersect the useful finite enclosures, retaining the Jensen lower bound; for small t, use convergent panel refinement instead. Negative Fourier coefficients reverse interval endpoints and are never projected to nonnegative numbers.

6 Exponential-fit Ram panels for rank one

The private proof suggests integrating the exponential exactly and expanding the log atom. We give the full resulting bound for the M004 rank-one observable. On a panel $J = [c-h/2, c+h/2] \subset [1, X]$, with $h > 0$ and $\rho = h/(2c) < 1$, define the polynomial

$$p_{q,c}(x) = \log c + \sum_{j=1}^{q} \frac{(-1)^{j+1}}{j} \left(\frac{x-c}{c}\right)^j \quad (q \geq 0),$$

and exact exponential moments $\mu_j(\beta, J) = \int_J e^{-\beta x}(x-c)^j \, \mathrm{d}x$. These are classical polynomial/exponential moments; odd moments can be signed, while $\mu_0 > 0$.

Theorem 6.1 (Panel error without exponential differentiation)**.** *For all real $\beta > 0$, integers $q \geq 0$, and the panels just specified,*

$$\left| \int_J e^{-\beta x} \log x \, \mathrm{d}x - \left(\log c \, \mu_0 + \sum_{j=1}^{q} \frac{(-1)^{j+1} \mu_j}{j c^j} \right) \right| \leq \frac{\rho^{q+1}}{(q+1)(1-\rho)} \mu_0(\beta, J). \tag{6.1}$$

If these panels partition $[1, X]$ and $g_M = \sum_{n=1}^{M} a_n e^{-\alpha n x}$, the error in approximating $2\alpha \int_1^X g_M(x) \log x \, \mathrm{d}x$ is at most

$$B_{\mathrm{panel}} = 2\alpha \sum_J \sum_{n=1}^{M} |a_n| \frac{\rho_J^{q_J+1}}{(q_J+1)(1-\rho_J)} \mu_0(\alpha n, J). \tag{6.2}$$

The omitted finite-series tail beyond $X > 1$ has the bound

$$B_x = 2 \sum_{n=1}^{M} \frac{|a_n|}{n} e^{-\alpha n X} \left(\log X + \frac{1}{\alpha n X} \right). \tag{6.3}$$

Consequently $B_1(M) + B_x + B_{\mathrm{panel}} + B_{\mathrm{round}}$ is a valid rank-one derivative error budget when B_{round} bounds all arithmetic evaluation errors.

Proof. For $z = (x-c)/c$ the absolutely convergent log series satisfies

$$\left| \log(1+z) - \sum_{j=1}^{q} (-1)^{j+1} z^j / j \right| \leq \sum_{j=q+1}^{\infty} |z|^j / j \leq \frac{\rho^{q+1}}{(q+1)(1-\rho)}.$$

Multiply by the positive measure $e^{-\beta x} \, \mathrm{d}x$ and integrate. The coefficient and panel sums follow by the triangle inequality; positivity of the newform is not used. Integration by parts gives

$$\int_X^{\infty} e^{-\beta x} \log x \, \mathrm{d}x = \frac{e^{-\beta X} \log X + \mathrm{E}_1(\beta X)}{\beta}.$$

Apply (5.5), multiply by $2\alpha|a_n|$, and add the independent Fourier tail of Theorem 5.1. □

Representation and conditioning. On each panel take $V_q = \mathcal{P}_q(J)$ with positive reference inner product $\int_J e^{-\beta x} uv \, \mathrm{d}x$. Its polynomial moment Gram matrix is positive definite, because a nonzero real polynomial cannot vanish almost everywhere on a nondegenerate interval. This is precisely the finite unisolvence premise read in private route P2. Centered or orthogonal coordinates can improve numerical conditioning, but do not change V_q or its approximation error. Recurrences for μ_j may suffer cancellation; an outward-rounded stable implementation is still required. The benefit of (6.1) is explicit: unlike a Taylor bound for $e^{-\beta x} \log x$, it never differentiates the rapidly decaying exponential and introduces no positive powers β^{q+1} in the remainder constant. This is not a uniform runtime comparison with modular symbols or mature approximate functional equations.

Termination boundary. For fixed E, M, X, refining panels so $\rho_J \to 0$ or increasing their degrees gives convergence; choosing M, X large controls the other tails. Thus arbitrary analytic accuracy is available in principle. An interval excluding zero proves nonvanishing. An interval containing zero does not prove vanishing, however small its width. In odd sign the exact central zero plus a nonzero derivative enclosure proves analytic rank one. In even sign a nonzero value proves analytic rank zero. These tests alone do not compute s_2 or an order of Sha.

7 Only the two-primary index is needed

Suppose now that the common rank $r \in \{0, 1\}$ is independently known. Let $L \subset \Lambda_E$ be a full-rank subgroup of finite index I and let R_L be its regulator using (2.2). In rank zero put $L = \Lambda_E = 0$, $I = R_L = 1$.

Theorem 7.1 (Dyadic saturation reduction)**.** *With $A_E > 0$ and $Q_E \in \mathbb{Q}_{>0}$ established independently, define $q_L = A_E(\#T_E)^2/(\Omega_E R_L C_E)$. Then*

$$R_L = I^2 \operatorname{Reg}_E, \qquad Q_E = I^2 q_L, \tag{7.1}$$

$$\textit{regulator correction: } \operatorname{Reg}_E = I^{-2} R_L, \qquad v_2(Q_E) = v_2(q_L) + 2v_2(I). \tag{7.2}$$

In particular, if L is two-saturated in Λ_E, meaning Λ_E/L has no element of order two, then $v_2(Q_E) = v_2(q_L)$. An uncomputed odd index is irrelevant to this valuation, although it still prevents recovering Q_E as an exact real or rational number.

Proof. In a full basis write the generators of L as the columns of an integer matrix M. The height matrix changes from G to $M^t G M$, and $I = |\det M|$, so its determinant gains I^2. The quotient divides by the regulator, proving both identities. A finite abelian group has no element of order two exactly when its order is odd, giving the last claim. Notice the opposite signs: the regulator's multiplicative index correction has valuation $-2v_2(I)$, whereas correction of the quotient has valuation $+2v_2(I)$. □

No two-adic valuation is being assigned to the individual real regulator: (7.2) concerns its rational multiplicative correction and the rational quotients. This distinction prevents mixing real numerical heights with dyadic arithmetic.

Corollary 7.2 (Rank-one halving certificate modulo torsion)**.** *Assume Λ_E has rank one and $P \in E(\mathbb{Q})$ is nontorsion. Suppose a complete arithmetic procedure can decide and solve*

$$[2]Q = P + T \quad \textit{for some } Q \in E(\mathbb{Q}),\ T \in T_E.$$

Repeatedly replace the class of P by a half while this is possible. If there are e successful halvings followed by a proved negative decision, the original index satisfies $v_2(I) = e$. Thus its height must be divided by 2^{2e} to remove the dyadic part of the index, and its quotient must be multiplied by 2^{2e} at the valuation level.

Proof. Choose an abstract generator G of Λ_E and write $[P] = mG$, $m \neq 0$. The equation modulo torsion is solvable exactly when m is even. Each successful half replaces m by $m/2$, and a negative decision means the remaining integer is odd. This takes exactly $v_2(m) = v_2(I)$ steps. The generator need not be known to justify the implication. □

Completeness of the rational halving test and of the torsion group is an arithmetic input, not an outcome of height positivity. Testing only $[2]Q = P$ can fail when a torsion shift is required. Likewise, being a local square or having a nonzero two-adic logarithm is not a global two-saturation certificate. The corollary narrows the private report's full-saturation requirement to the actual prime-specific task; it does not supply a Mordell–Weil rank algorithm.

7.1 Conversion from a declared alternative implementation

If an implementation deliberately uses the connected period Ω_0 and a pairing $\widetilde{B} = B/\kappa$, for a known rational $\kappa > 0$, let $\widetilde{R}_L = R_L/\kappa^r$ and

$$q_{\mathrm{raw}} = \frac{A_E(\#T_E)^2}{\Omega_0 \widetilde{R}_L C_E}.$$

The exact conversion, not an accusation of source misnormalization, is

$$Q_E = \frac{I^2}{c_\infty \kappa^r} q_{\mathrm{raw}}, \qquad v_2(Q_E) = v_2(q_{\mathrm{raw}}) + 2v_2(I) - v_2(c_\infty) - r v_2(\kappa). \tag{7.3}$$

This follows directly by substituting the definitions. Using all real components from the outset sets the separate period correction to one. Using exactly the source pairing sets $\kappa = 1$. At rank zero the height-convention exponent is zero. These local formulas are sufficient for the report and do not duplicate the parent's cross-report checklist.

8 A signed determinant-defect extension

The most source-specific local deduction retains a defect which is zero in the source's integral-unit conclusion. We state its arithmetic prerequisites explicitly rather than suggesting that analytic panels prove them.

Work first in the reducible setting actually read in source files 08–09: $E/\mathbb{Q}$ is non-CM with rational two-torsion, $K = \mathbb{Q}(\sqrt{-d})$ for a prime $d \equiv 7 \pmod{16}$, and every prime dividing $2N$ splits in K. Put $k = -d$. Assume $L(E/K, s) = L(E, s)L(E^k, s)$ has a simple zero at one. Import the classical finite-Sha and split Gross–Zagier/isogeny comparisons in source file 02. Let $P \in E(K)$ be the specified single Hilbert-class trace, not a multiple or an averaged trace. It is nontorsion in this setting. Write

$$n_P = v_2([E(K)/E(K)_{\mathrm{tors}} : \mathbb{Z}\overline{P}]), \quad \tau_g = v_2(\#E(K)_{\mathrm{tors}}), \quad s_K = v_2(\#\,\mathrm{Sha}(E/K)).$$

For the modular parametrization $\phi^*\omega_E = c_E f\, \mathrm{d}q/q$, put $m_E = v_2(c_E)$. The full group index includes torsion:

$$v_2([E(K) : \mathbb{Z}P]) = n_P + \tau_g. \tag{8.1}$$

Indeed the quotient maps to the free quotient, and its kernel has order $\#E(K)_{\mathrm{tors}}$ since $\mathbb{Z}P$ intersects torsion trivially.

Let $D = L_{\mathrm{alg}}(C)(0)$ and $B_0 = B(0)$ be nonzero central determinant and paired-measure scalars in a common dyadic field. To avoid confusion, B_0 here is not the tail bound $B_0(M)$. For a ramified scalar extension the valuation is still normalized by $v_2(2) = 1$, not by $v(\pi) = 1$ without conversion. Set

$$z = v_2(\log_{\omega_E} P), \qquad \log_{\omega_E} E(\mathbb{Q}_2) = 2^l\mathbb{Z}_2, \qquad p_\ell = v_2(P_\ell(1)), \quad P_\ell(1) = L_\ell(E, 1)^{-1}.$$

The nonzero local logarithm and the source's actual integral local lattices are required. For clarity, take the central support to be $2N$; extra split good support primes contribute cancelling terms by the same Haar identities.

Source input: The following are the central comparison identities read in file 09, lines 398–427, before using the unit conclusion:

$$-v_2(D) = 2n_P + 2\tau_g - s_K - 2(z - l + \tau_2) - 2\sum_{\substack{q|N \\ q\neq 2}} \tau_q, \tag{8.2}$$

$$v_2(B_0) = -2m_E + 2z + 2\sum_{\ell|2N} p_\ell, \tag{8.3}$$

$$\tau_2 - l = v_2(c_2) + p_2, \qquad \tau_q = v_2(c_q) + p_q \quad (q \neq 2). \tag{8.4}$$

The first uses strict/finite comparison and the Kummer volume. The second is the exact paired logarithmic interpolation at the center. The third uses minimal-differential Haar measures. The construction and integrality of these identities are imported, including their two-place multiplicities and the logarithmic lattice at two.

Theorem 8.1 (Central defect propagation)**.** *Under the preceding hypotheses and identities, define* $\mu = v_2(D/B_0)$*. Then*

$$\mu = s_K + 2m_E + 2\sum_{\ell|N} v_2(c_\ell) - 2(n_P + \tau_g), \tag{8.5}$$

$$X(E) + X(E^k) = -\mu. \tag{8.6}$$

Thus the unit case is exactly $\mu = 0$*. More generally a certified bound* $a \leq \mu \leq b$ *gives* $-b \leq X(E) + X(E^k) \leq -a$*. For this specified arithmetic situation* μ *is an integer even if the chosen coefficient field is ramified.*

Proof. Negate (8.2) and subtract (8.3). The two z terms cancel, giving

$$\mu = s_K + 2m_E - 2(n_P + \tau_g) + 2(\tau_2 - l) + 2\sum_{\substack{q|N \\ q\neq 2}} \tau_q - 2\sum_{\ell|2N} p_\ell.$$

Substitute (8.4); every Euler factor cancels, proving (8.5). Source file 02's full-index identity is

$$X(E) + X(E^k) = v_2\left(\frac{[E(K) : \mathbb{Z}P]^2}{c_E^2 \prod_\ell c_\ell(E)^2 \, \#\,\mathrm{Sha}(E/K)}\right).$$

Use (8.1) to obtain (8.6). All terms of (8.5) are integers, proving the integrality assertion. Reversing an interval under multiplication by -1 gives the final bound. □

Corollary 8.2 (Known-rank integrality suffices for the final valuation)**.** *In the setting of Theorem 8.1, assume additionally the source's nonnegativity* $X(E), X(E^k) \geq 0$*. If the nonzero scalar* D/B_0 *is integral at two, then* $\mu = 0$ *and* $X(E) = X(E^k) = 0$*.*

Proof. Integrality gives $\mu \geq 0$. Equation (8.6) and nonnegativity give $-\mu \geq 0$. Hence $\mu = 0$, and two nonnegative summands with sum zero both vanish. □

This is a limited weakening of the input needed *after* a simple analytic zero and all central identities are known. It is not a proof that the source's unit construction can be omitted at an unknown Selmer-corank-one center: source file 09 uses that unit first to show $Z(0) \neq 0$ and hence prove analytic rank. It is also not an assertion that the numerator/denominator identities automatically hold in every residual branch. They can be reused in another branch only after its own comparison identities have been established. A scalar unit at this center is not a claim of a globally primitive multivariable series in an unspecified coefficient ring.

Quantitative error accounting. An unknown integral clearing multiplier of valuation e applied to a point contributes $2e$ through its height or index square. A multiplier of a paired determinant likewise has its stated multiplicity. Horizontal identities after inverting two cannot determine these missing terms. Equation (8.5) is useful precisely because it leaves all such contributions visible. If a proposed computation yields $\mu > 0$ while also claiming the known-rank nonnegativity premises, the inputs are inconsistent; rounding should not be used to erase the discrepancy.

9 From real intervals to dyadic conclusions

Suppose independently certified positive intervals are available for A_E, Ω_E, and the subgroup regulator R_L:

$$A_E \in [a_-, a_+], \quad \Omega_E \in [o_-, o_+], \quad R_L \in [h_-, h_+], \qquad a_-, o_-, h_- > 0.$$

With exact torsion and Tamagawa factors, interval monotonicity gives

$$q_L \in J = \left[\frac{a_-(\#T_E)^2}{o_+ h_+ C_E}, \frac{a_+(\#T_E)^2}{o_- h_- C_E} \right]. \tag{9.1}$$

In rank zero $h_- = h_+ = 1$; in rank one $R_L = H(P)$. Positive height certifies nontorsion but does not certify saturation. Source rationality gives $q_L \in \mathbb{Q}_{>0}$ once the relevant rank premises hold. Rationality alone is insufficient for the next step.

Proposition 9.1 (A valuation cannot be read from real proximity)**.** *For every integer e, the positive rationals of valuation $v_2 = e$ are dense in $\mathbb{R}_{>0}$. Hence no real interval with nonempty interior determines the valuation of an otherwise unrestricted positive rational.*

Proof. For any positive target x, choose arbitrarily large positive odd denominators b. Choose a positive odd numerator a nearest to $2^{-e}xb$; for sufficiently large b, $|a - 2^{-e}xb| \leq 1$. Then $2^e a/b$ has valuation e and distance at most $2^e/b$ from x. For fixed e this tends to zero. Reduction of the odd numerator and denominator does not change the valuation. □

Theorem 9.2 (Conditional denominator and valuation isolation)**.** *Let $J \subset \mathbb{R}_{>0}$ be a bounded closed interval and let $q_L \in J$ be rational.*

1. *If an independent arithmetic certificate gives $Dq_L \in \mathbb{Z}$ for a known integer $D \geq 1$, and DJ contains a unique integer n, then $q_L = n/D$ and $v_2(Q_E) = v_2(n) - v_2(D) + 2v_2(I)$.*
2. *If instead the reduced denominator of q_L is independently bounded by $B \geq 1$, an interval of width strictly less than $1/B^2$ contains at most one such rational. Because q_L is known to belong, finding that candidate determines q_L and the same index correction.*

3. *At larger widths, enumerate the finite set of positive reduced fractions in J with denominator at most B. If every candidate has the same valuation e, the conclusion $v_2(Q_E) = e + 2v_2(I)$ is valid even without determining q_L uniquely.*

Proof. The first assertion is integer isolation. Distinct reduced rationals $a/b, c/d$ with $b, d \le B$ differ by $|ad - bc|/(bd) \ge 1/(bd) \ge 1/B^2$, proving the second. On a bounded interval there are finitely many possible numerators for each of the finitely many denominators. The true value is one of those candidates, so unanimity of their valuations proves the third assertion. □

The independent denominator premise is deliberately strong. This report does not supply a universal denominator bound for q_L in rank one, nor derive it from BSD. It can come from separately certified modular symbols or an applicable arithmetic comparison with all indices controlled. A bound only on the *two-part* of the denominator does not suffice for real isolation, because odd denominators remain unbounded.

Even if the source's $v_2(Q_E) = v_2(\#\,\mathrm{Sha})$ is accepted, that equality does not make Q_E a positive integer or a square in $\mathbb{Q}$. Alternation of a finite elliptic Sha gives additional arithmetic constraints on its order, but imposing those constraints on an analytic quotient to prove the very equality in question would be circular.

10 Finite checks and proof-producing workflow

The report-specific test script checks exact rational index conversions, two-saturation halving in abstract rank-one lattices, the signed central-defect cancellation, rational separation, and counterexamples to proximity-based valuation. Bounded high-precision tests also compare the exponential-integral identities, signed finite remainders, and panel-error estimates with numerical integration. The latter are regression checks, not rigorous outward-rounded elliptic-curve data. The identities and inequalities are justified by the written proofs, not by the finite test count. No Selmer group, actual Heegner trace, integral cochain system, or full Sha was computed by this author.

An exact manufactured defect example. Take valuations $n_P = 3$, $\tau_g = 1$, $s_K = 2$, $m_E = 1$, and $\sum v_2(c_\ell) = 2$. Equation (8.5) gives $\mu = 0$. Increasing s_K to 4 while keeping the other data gives $\mu = 2$ and total discrepancy -2. This manufactured second case cannot simultaneously satisfy the source's nonnegativity premises; it is an intentional inconsistency detector, not an asserted elliptic-curve example.

An exact index example. For a rank-one lattice with full height $H(G) = 5$ and $P = 6G$, the displayed height is 180. The full regulator is $180/36 = 5$. If $q_L = 3/20$, the full quotient is $27/5$, with valuation 0; $v_2(3/20) = -2$ and $2v_2(6) = 2$. Halving once replaces P by $3G$, leaving an odd index three and the same two-primary quotient valuation. The unresolved odd index still changes the exact quotient by nine.

A fail-closed workflow.

1. Fix the exact minimal model, conductor, root number, torsion, finite local factors, differential, and the chosen height convention. Supply rank or Selmer premises by identified arithmetic results, not by a small numerical leading term.

2. Compute the appropriate low-rank kernel enclosure, retaining Fourier tail, kernel/panel error and arithmetic rounding separately. Reject rank certification whenever the needed interval contains zero.

3. For rank one, certify a positive height and the two-part of its index using a complete arithmetic procedure. An unresolved odd index can remain only for the valuation task.

4. Propagate positive intervals by (9.1). Request exact rational arithmetic or the independent denominator premise of Theorem 9.2; otherwise report only an interval.

5. In a split comparison, record the three central identities and their actual lattice and trace normalizations before evaluating μ. Apply the defect theorem only after its rank and local premises.

6. Keep three distinct outcomes: analytic nonvanishing certified; arithmetic valuation certified conditionally on named inputs; or insufficient information. None is relabeled a new universal BSD proof.

11 Advantages, limitations, and further research

Advantages. The low-rank kernels avoid unstable numerical differentiation and eliminate spatial quadrature altogether in rank zero. The rank-one exponential-fit panels avoid differentiating large exponential rates. The dyadic saturation reduction states exactly what index information the two-primary task needs. The signed defect identity turns a missing unit assertion into an explicit arithmetic quantity rather than an unspecified error. Each benefit addresses an actual M004 normalization or comparison, and the formulas remain usable without treating signed modular coefficients as a positive measure.

Limitations and alternatives. The conductor makes α small and can force long coefficient lists; there is no family-uniform efficiency theorem here. Standard approximate functional equations already implement the underlying analytic ideas. Modular symbols can be preferable in rank zero because they supply exact rational information, not just intervals. Descent, isogenies, Heegner constructions and Iwasawa theory supply arithmetic information that Ram-coordinate changes do not. Height conditioning matters in higher rank, but M004 needs only a scalar height in rank one; the private higher-rank determinant bounds are not credited as a new improvement here.

The deep imported boundary includes the integral Siegel-unit lattice and full-level reciprocity, residual concentration, exact finite-model and limit compatibility, tame spectral/intersection identity, split local logarithm lattices and paired measures, elliptic-unit and ray-unit primitivity, integral conjugate alternation, and the real Hecke/Pfaffian construction. The real detector proof was read selectively in context; that does not verify the later integral cofactor lift. The parent's identity and normalization checks do not close any of these proof tasks.

Specific next problems. First, implement the exponential-fit panels with directed rounding and compare total certified work, at fixed error, against a trusted approximate-functional-equation implementation across conductor ranges. Record the exact curve equation, not an ambiguous abbreviated label. Second, couple a complete torsion-aware halving solver to the scalar height certificate and demonstrate that discarding odd saturation work actually saves computation on rank-one examples. Third, identify explicit denominator bounds for normalized rank-one quotients from independent arithmetic comparisons; without these, interval-based valuation extraction remains

conditional. Fourth, formalize the central determinant calculation over an explicitly normalized DVR, including torsion in the full index and the two split-place volumes. Finally, investigate whether integrality plus independent analytic nonvanishing is available in concrete families where a full multivariable unit theorem is inconvenient. That would make Corollary 8.2 useful, but is a research task, not an achievement claimed here.

Final assessment. The realized contribution is C2/5, independently of applicability L3. The report proves modest analytic and arithmetic-certification refinements and an explicit conditional defect extension. It neither improves the source's universal range nor verifies the entire arithmetic proof. Successful compilation, page inspection, hashes and bounded tests support reproducibility and presentation, not mathematical priority or the truth of every imported theorem.

A Evidence record and reproducibility boundaries

The local audit files provide the literal-read intervals, pre/post-read SHA-256 endpoints, whole structure map, actual test results, public refresh scopes, native compile attempt, exported build diagnostics, all-page visual review, contribution assessment, and final freeze. The 17 source and two private files were hashed before their scientific reads and again at finalization. Equal endpoints do not constitute continuous change monitoring. The administrative project README and M719 audit examples establish the evidence format; their earlier-stage exclusion of L3 does not override the user's explicit new authorization.

One combined public-search response was truncated, but its separately returned 203-line introduction read was complete. No source/private literal interval in the ledger was truncated. Public claims used here are supported by separately opened primary records or, for LTYZ, the explicitly limited primary search abstract. No complete external literature proof was read. The external source's separate any-prime Selmer converse and signed-squarefree density companion are imports.

The final manifest excludes mutable shared queue, progress and catalogue files, other manuscripts, and its own self-hash. The report-specific snapshot of parent evidence is frozen; the shared parent files are not modified. The source map is structural evidence, not extra lines added to the literal-read count. Numerical tests using high precision are marked separately from exact rational tests and are not interval-certified number-theoretic computations.

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